Consider uniform configuration of $\varphi(\vec{r}) = \varphi$ first:

$$S_{LG} = \underset{\int d\vec{r}}{V}\left(\underset{\frac{1}{2}m^2}{A}\varphi^2 + \underset{\frac{1}{4}u}{B}\varphi^4 + \dots \right)$$

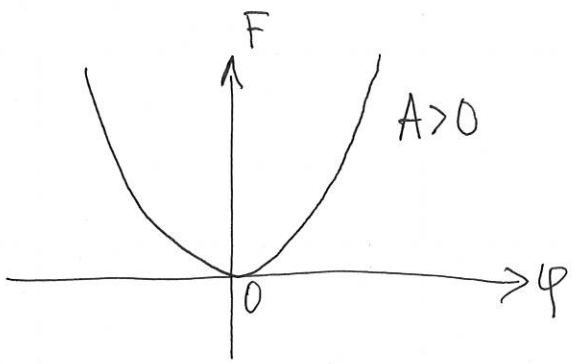
Landau's free energy

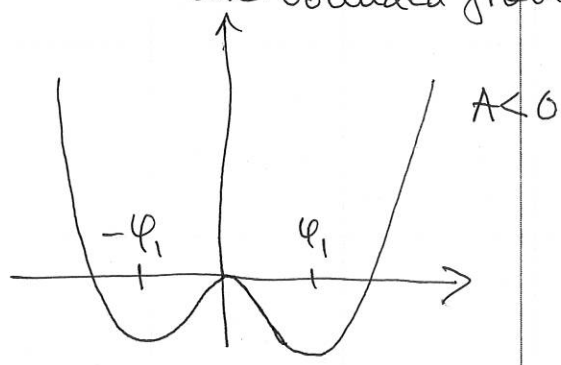
φ = order parameter

$$Z = \exp(-\beta F) \Rightarrow F = -\frac{1}{\beta}\left(-S_{LG}\right) = \underset{\uparrow}{T}\, V\left(A\varphi^2 + B\varphi^4 + \dots\right)$$

can replace by T_c (= const) since this is ~~where most of development~~ interesting regime

<u>What is φ?</u> $B > 0$ for stability (so that S_{LG}, F are bounded from above)





$$\frac{\partial F}{\partial \varphi} = 0 = 2A\varphi + 4B\varphi^3 = 2\varphi(A + 2B\varphi^2)$$

1) $A > 0 \Rightarrow \varphi = 0 \Rightarrow F$ has min at $\varphi_0 = 0$, $F(\varphi_0) = 0$.

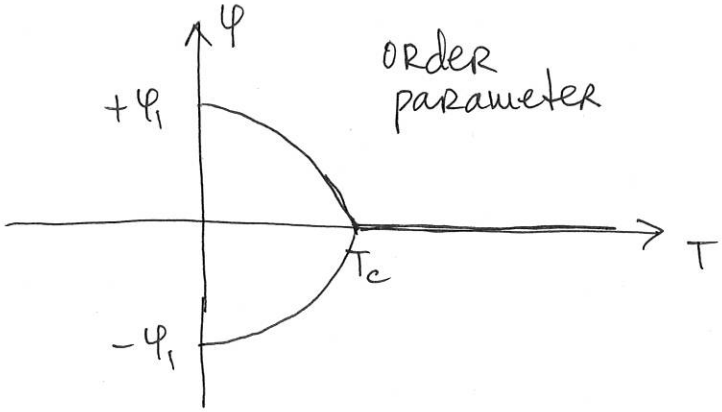
2) $A < 0$, $A = -|A| \Rightarrow \varphi_0 = 0$ <u>OR</u> $2B\varphi_1^2 = -A = |A|$

$$\varphi_1 = \pm\sqrt{\frac{|A|}{2B}} = \pm\sqrt{\frac{-A}{2B}}$$

$$F(\varphi_0) = 0 \quad ; \quad F(\varphi_1) = T_c V\left(-|A|\varphi_1^2 + B\varphi_1^4\right) =$$

$$= T_c V\left\{-|A|\frac{|A|}{2B} + B\left(\frac{|A|}{2B}\right)^2\right\} = T_c V\left\{-\frac{A^2}{4B}\right\} < F(\varphi_0)$$

Thus, at $A=0$ we have continuous phase transition at $T=T_c$



$$A = \frac{1}{2} m^2 = \frac{T-T_c}{2T_c}$$

$$\varphi_1 = \pm\sqrt{\frac{T_c - T}{4BT_c}}$$

$$\varphi_1 = \pm\sqrt{\frac{-\frac{1}{2}m^2}{\frac{2}{4}u}} = \pm\sqrt{\frac{-m^2}{u}}$$

But which of 2 allowed (and degenerate) solutions ($+\varphi_1$ or $-\varphi_1$) is realized? $\longrightarrow$ $\boxed{\text{Spontaneous symmetry breaking}}$

$\Uparrow$

decided randomly

Below T_c:

$$S_{LG} = \int d\vec{r} \left(\frac{1}{2}\rho (\partial_r \varphi)^2 + \frac{1}{2} m^2 \varphi^2 + \frac{1}{4} u \varphi^4 \right)$$

$$m^2 = \frac{T-T_c}{T_c} < 0 \quad , \quad m^2 = -\mu^2$$

Assume $+\varphi_1$ is chosen : $\varphi = \varphi_1 + \delta\varphi$

$$S_{LG} = \int d\vec{r} \; \frac{1}{2}\rho (\partial_r \delta\varphi)^2 - \frac{1}{2}\mu^2 (\varphi_1 + \delta\varphi)^2 + \frac{1}{4} u (\varphi_1 + \delta\varphi)^4$$

$$= \int d\vec{r} \left\{ \frac{1}{2}\rho (\vec{\nabla}\delta\varphi)^2 - \frac{1}{2}\mu^2 \left(\varphi_1^2 + \underline{2\varphi_1 \delta\varphi} + \underline{\delta\varphi^2} \right) \right.$$

$$\left. + \frac{1}{4} u \left(\varphi_1^4 + (\delta\varphi)^4 + \underline{4\varphi_1^3 \delta\varphi} + \underline{6\varphi_1^2 \delta\varphi^2} + 4\varphi_1 \delta\varphi^3 \right) \right\}$$

$$= \int d\vec{r} \; \frac{1}{2}\rho (\vec{\nabla}\delta\varphi)^2 \underbrace{- \frac{1}{2}\mu^2 \varphi_1^2 + \frac{1}{4} u \varphi_1^4}_{\text{constant}} + \delta\varphi \underbrace{\left(-\mu^2 \varphi_1 + u \varphi_1^3 \right)}_{0 \text{ due to minimization}} +$$

continued $+ \delta\varphi^2 \underbrace{\left(-\frac{1}{2}\mu^2 + \frac{6}{4}\varphi_1^2 u\right)}_{\frac{1}{2}\left(-\mu^2 + 3u\left(-\frac{m^2}{u}\right)\right) = \frac{1}{2}\left(-\mu^2 + 3\mu^2\right) = \mu^2} + u\varphi_1 \delta\varphi^3 + \frac{1}{4} u \delta\varphi^4 \Big\}$

$T<T_c$ $$S_{LG} = \int d\vec{r} \left(\frac{1}{2}\rho \left(\nabla \delta\varphi\right)^2 + \mu^2 \delta\varphi^2 + \underline{\underline{u\varphi_1 \delta\varphi^3}} + \frac{1}{4} u \delta\varphi^4 \right)$$

broken symmetry

$T>T_c$ $$S_{LG} = \int d\vec{r} \left(\frac{1}{2}\rho \left(\nabla \varphi\right)^2 + \frac{1}{2} m^2 \varphi^2 + \frac{1}{4} u \varphi^4 \right)$$

symmetric, $S_{LG}(\varphi) = S_{LG}(-\varphi)$.

$S_{LG}(\delta\varphi) \neq S_{LG}(-\delta\varphi)$

First vs Second order transitions

1. $S_{LG} = \int d\vec{r} \left(\frac{1}{2} \rho (\vec{\nabla}\varphi)^2 + \frac{1}{2} m^2 \varphi^2 + \frac{1}{4} u \varphi^4 \right)$

2. Consider generalization

$$S = \int d\vec{r} \left(\frac{1}{2} \rho (\vec{\nabla}\varphi)^2 + \frac{1}{2} m^2 \varphi^2 + \frac{1}{4} u \varphi^4 + \frac{1}{6} v \varphi^6 \right)$$

where $u > 0$ or $u < 0$, but $v > 0$ (for stability)

Here, all configurations are uniform $\varphi(\vec{r}) = \varphi$.

$$S = V F(\varphi) = V \left(\frac{1}{2} m^2 \varphi^2 + \frac{1}{4} u \varphi^4 + \frac{1}{6} v \varphi^6 \right)$$

let $m^2 = t$, so that $t < 0$ or $t > 0$ naturally.

Analyze extrema: $\frac{\partial F}{\partial \varphi} = 0 = t\varphi + u\varphi^3 + v\varphi^5$

$\varphi = 0$ OR $t + u\varphi^2 + v\varphi^4 = 0$

$F(0) = 0$

let $x = \varphi^2$: $v x^2 + u x + t = 0$

$$\varphi^2 = x = \frac{-u \pm \sqrt{u^2 - 4vt}}{2v}$$

$\Rightarrow$ two other minima are possible provided $u < 0$, so that $-u = |u|$ and $u^2 - 4vt > 0 \Rightarrow t < \frac{u^2}{4v}$.

Denote them $\pm \varphi_1$, where

$$\varphi_1^2 = \frac{|u|}{2v} + \frac{\sqrt{u^2 - 4vt}}{2v}.$$

Note that

$$\varphi_2^2 = \frac{|u|}{2v} - \frac{\sqrt{u^2 - 4vt}}{2v}$$ describes <u>maximum</u>

Study next $\frac{\partial^2 F}{\partial \varphi^2} = t + 3u\varphi^2 + 5v\varphi^4$

$$\frac{\partial^2 F}{\partial \varphi^2} = t \left(\text{for } \varphi_0 = 0\right) > 0 \ \underline{\text{for } t > 0}$$

$\Rightarrow$ minimum for $\underline{t>0}$

$$\left.\frac{\partial^2 F}{\partial \varphi^2}\right|_{\varphi_{1,2}^2} = t + \frac{3u}{2v}\left(-u \pm \sqrt{u^2-4vt}\right) + \frac{5v}{(2v)^2}\left(-u \pm \sqrt{u^2-4vt}\right)^2$$

$$= t - \frac{3u^2}{2v} \pm \frac{3u}{2v}\sqrt{u^2-4vt} + \frac{5}{4v}\left(u^2 + u^2 - 4vt \mp 2u\sqrt{u^2-4vt}\right) =$$

$$= t + \frac{u^2}{u} - 5t \pm \sqrt{u^2-4vt}\left(\frac{3u}{2v} - \frac{5u}{2v}\right)$$

$$= \frac{u^2}{v} - 4t \pm \sqrt{u^2-4vt}\left(-\frac{u}{v}\right) =$$

$$= \frac{1}{v}\left(\sqrt{u^2-4vt} \mp u\right)\sqrt{u^2-4vt} = \frac{\sqrt{u^2-4vt}}{v}\left(\sqrt{u^2-4vt} \mp u\right)$$

$$= \frac{\sqrt{u^2-4vt}}{v}\left(\sqrt{u^2-4vt} \pm |u|\right)$$

$$\Rightarrow \left.\frac{\partial^2 F}{\partial \varphi^2}\right|_{\varphi_1^2} > 0 \Rightarrow \underline{\text{minimum}}$$

$$\left.\frac{\partial^2 F}{\partial \varphi^2}\right|_{\varphi_2^2} < 0 \Rightarrow \text{maximum}$$

$$F(\varphi_1^2) = \frac{1}{2} t \varphi_1^2 + \frac{1}{4} u \varphi_1^4 + \frac{1}{6} v \varphi_1^6 =$$

$$= \frac{1}{2} \varphi_1^2 \left(t + \frac{1}{2} u \varphi_1^2 + \frac{1}{3} v \varphi_1^4 \right) = \frac{|u| + \sqrt{u^2 - 4vt}}{4v} \cdot$$

$$\cdot \left(t - \frac{|u|}{2} \frac{|u| + \sqrt{u^2 - 4vt}}{2v} + \frac{v}{3} \frac{\left(|u| + \sqrt{u^2 - 4vt}\right)^2}{4v^2} \right) =$$

$$= \frac{|u| + \sqrt{u^2 - 4vt}}{4v} \cdot \left(t - \frac{u^2}{4v} - \frac{|u|\sqrt{u^2 - 4vt}}{4v} + \frac{u^2 + u^2 - 4vt + 2|u|\sqrt{}}{12v} \right)$$

$$= \frac{|u| + \sqrt{u^2 - 4vt}}{4v} \underbrace{\left(t + \frac{-u^2 - 4vt - |u|\sqrt{u^2 - 4vt}}{12v} \right)}_{t - \frac{u^2 + 4vt + |u|\sqrt{u^2 - 4vt}}{12v}}$$

$$\Rightarrow F(\varphi_1^2) = 0 \quad \text{for } t_1 = \frac{1}{8v} \left(u^2 + |u|\sqrt{u^2 - 4vt_1} \right)$$

$$\left(8vt_1 - u^2\right)^2 = \left(|u|\sqrt{u^2 - 4vt_1}\right)^2 = u^2 \left(u^2 - 4vt_1\right)$$

$$64v^2 t_1^2 - 16vu^2 t_1 + \cancel{u^4} = \cancel{u^4} - 4vu^2 t_1$$

$$64v^2 t_1^2 = 12vu^2 t_1$$

$$\Rightarrow t_1 = \frac{12u^2}{64v} = \frac{3u^2}{16v}$$

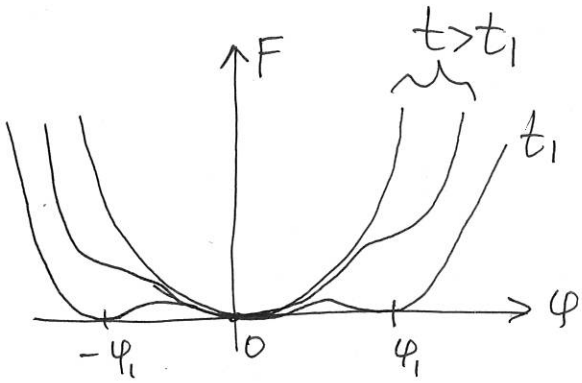
$\Rightarrow$ temperature $t_1 = \frac{3u^2}{16v}$ $\quad F(\varphi_1^2) = 0 = F(\varphi_0 = 0)$

So two values of the order parameter, $\varphi_0 = 0$ and

and $\varphi_1^2 = \frac{|u| + \sqrt{u^2 - 4vt_1}}{2v} = \frac{|u| + \sqrt{\frac{u^2}{4}}}{2v} = \frac{3|u|}{4v}$,

are degenerate!

$$4vt_1 = \frac{3u^2}{4}$$

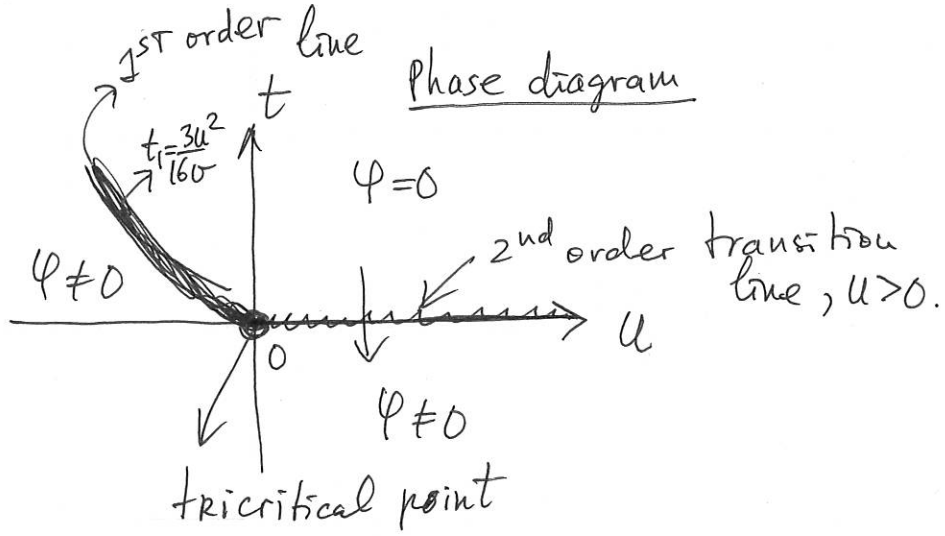
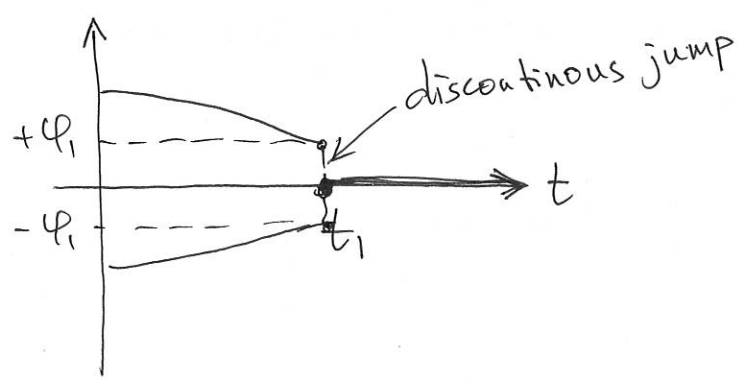


At $t<t_1$, the minima at $\varphi_1^2 = \dfrac{|u|+\sqrt{u^2-4vt}}{2v}$ become <u>lower</u> than that at $\varphi_0=0$

$\Rightarrow$ order parameter changes <u>discontinuously</u>

at $t_1 = \dfrac{3|u|}{4v} > 0 \Rightarrow$ this is <u>first order transition</u> !

<u>Phase diagram</u>



ORDER parameter $\varphi_1 = \sqrt{\dfrac{3|u|}{4v}}$ at the transition, $t=t_1$.